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### FOLDABLE CRUTCH STOOL

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#### Abstract

The present invention discloses a foldable crutch stool, including: a support assembly, including an upper support rod and a lower support rod, which are telescopic; a collapsing assembly; and a stool assembly with an upper end pivoted to a sliding base and a lower end pivoted to a positioning base via collapsible rods, wherein a bottom surface of a lower end of the stool assembly is provided with first snap-fit portions, the positioning base is provided with a connector which is provided with second snap-fit portions, and when the stool assembly is folded, the first snap-fit portions are snap-fitted with the second snap-fit portions.

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#### Background/Summary

# BACKGROUND OF THE INVENTION

## Technical Field

[0001] The present invention relates to the technical field of crutch stools, and in particular to a foldable crutch stool.

## Description of Related Art

[0002] In the related art, a crutch stool generally includes a support rod and a foldable stool assembly connected to the support rod via a positioning base and a sliding base. However, when the stool assembly is folded, the connection between the stool assembly and the support rod or the positioning base is not steady, such that the stool assembly in the folded state is still likely to be unfolded accidentally, which affects the user's use experience. Furthermore, the accidental unfolding of the stool assembly is likely to cause accidents, resulting in a potential safety hazard.

[0003] Therefore, it is necessary to make improvements to the prior art.

## BRIEF SUMMARY OF THE INVENTION

[0004] The present invention is intended to solve one of the problems existing in the related art to some extent. To this end, the present invention provides a foldable crutch stool, with a stool assembly that has a reliable connection with a positioning base when in a folded state and is unlikely to be unfolded accidentally.

[0005] The above object is achieved by technical solutions as follows: [0006] a foldable crutch stool, including: [0007] a support assembly, including an axially disposed support rod, wherein the support rod includes an upper support rod and a lower support rod, the lower support rod is provided with a first regulating hole close to an upper end of the lower support rod, and the upper support rod is vertically movably disposed in the lower support rod, and is provided with several second regulating holes mated with the first regulating hole; [0008] a connecting member sleeved outside the lower support rod for connecting the first regulating hole and the second regulating hole corresponding thereto in position, together; [0009] a handle assembly disposed on a top of the upper support rod; [0010] a collapsing assembly, including a positioning base, a sliding base and collapsible rods, wherein the positioning base is sleeved on the lower support rod, and the sliding base is axially slidably sleeved on the lower support rod and above the positioning base; and [0011] a stool assembly with an upper end pivoted to the sliding base and a lower end pivoted to the positioning base via the collapsible rods, wherein a bottom surface of a lower end of the stool assembly is provided with first snap-fit portions, the positioning base is provided with a connector which is provided with second snap-fit portions, and when the stool assembly is folded, the first snap-fit portions are snap-fitted with the second snap-fit portions.

[0012] In some embodiments, the stool assembly includes a middle seat plate and side seat plates disposed on both sides of the middle seat plate, the side seat plates are connected to the middle seat plate in a foldable way, and bottom surfaces on lower ends of the middle seat plate and the side seat plates are provided with the first snap-fit portions.

[0013] In some embodiments, the connector is provided with three second snap-fit portions to correspondingly mate three first snap-fit portions on the stool assembly.

[0014] In some embodiments, the connector is axially slidably sleeved on the lower support rod and between the positioning base and the sliding base, the connector is provided with connecting rods connected to the positioning base, and springs are sleeved outside the connecting rods and sandwiched between the connector and the positioning base.

[0015] In some embodiments, two connecting rods are provided and located on either side of the connector respectively, and lower ends of the connecting rods are provided with hooks for connection with the positioning base.

[0016] In some embodiments, the positioning base is provided with a connecting base at bottom, a lower end of the stool assembly is pivoted to the connecting base via the collapsible rods.

[0017] In some embodiments, the connecting member includes an elastic snap ring and a

connecting pin disposed on an inner side of the elastic snap ring, the elastic snap ring is sleeveable outside the lower support rod, the connecting pin is inserted into the first regulating hole and the corresponding second regulating hole, and the elastic snap ring is provided with a press portion. [0018] In some embodiments, the lower support rod and the upper support rod are both polygonal in cross section, the handle assembly is 7-shaped and includes a transversely disposed hand gripping portion and a connecting portion extending from a bottom of the hand gripping portion, the connecting portion is inserted into the upper support rod to achieve connection, a peripheral surface of the connecting portion is provided with reinforcing ribs that are circumferentially distributed at interval, and the reinforcing ribs are axially arranged.

[0019] In some embodiments, a decorative ring is sleeved on a top of the lower support rod, and the upper support rod is inserted into the decorative ring.

[0020] In some embodiments, the lower support rod is provided with at least two foldable support legs, which are provided with anti-skid pads at bottom.

[0021] Compared with the prior art, the present invention has at least the following beneficial effects: [0022] 1. according to the foldable crutch stool provided by the present invention, the stool assembly in a folded state is reliably connected to the positioning base, such that the stool assembly in the folded state is unlikely to be unfolded accidentally, which can improve the users' use experience, avoid accidents caused by accidentally unfolding the stool assembly, and facilitate improvement of product safety; and [0023] 2. the foldable crutch stool provided by the present invention is adjustable in height and convenient to operate and use.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0024] FIG. 1 is a first schematic stereoscopic view of a crutch stool in an unfolded state according to an embodiment;

[0025] FIG. 2 is a second schematic stereoscopic view of the crutch stool in the unfolded state according to an embodiment;

[0026] FIG. 3 is a partially enlarged view of a section A in FIG. 2;

[0027] FIG. 4 is a schematic stereoscopic view of the crutch stool in a folded state according to an embodiment;

[0028] FIG. 5 is a schematic stereoscopic view of the crutch stool according to an embodiment;

[0029] FIG. 6 is a partially exploded view of the crutch stool according to an embodiment;

[0030] FIG. 7 is a partially enlarged view of a section B in FIG. 6;

[0031] FIG. 8 is a schematic stereoscopic view of a connecting member according to an embodiment; and

[0032] FIG. 9 is a schematic stereoscopic view of a handle assembly according to an embodiment.

### DETAILED DESCRIPTION OF THE INVENTION

[0033] The present invention is explained in the embodiments below, but is not limited by these embodiments. The modifications to the specific embodiments of the present invention and the equivalent substitution of some of the technical features shall be included within the scope of the technical solutions claimed by the present invention.

[0034] Embodiment I: As shown in FIGS. 1, 2, 3, 4, 5, 6, 7, 8 and 9, this embodiment provides a foldable crutch stool, which includes: [0035] a support assembly 1, including an axially disposed support rod 11, wherein the support rod 11 includes an upper support rod 101 and a lower support rod 102, the lower support rod 101 is provided with a first regulating hole 111 close to an upper end of the lower support rod, and the upper support rod 102 is vertically movably disposed in the lower support rod 101, and is provided with several second regulating holes 121 mated with the first regulating hole; [0036] a connecting member 103 sleeved outside the lower support rod 101 for

connecting the first regulating hole **111** and the second regulating hole **121** corresponding thereto in position, together; [0037] a handle assembly **104** disposed on a top of the upper support rod **102**; [0038] a collapsing assembly **2**, including a positioning base **21**, a sliding base **22** and collapsible rods **23**, wherein the positioning base **21** is sleeved on the lower support rod **101**, and the sliding base **22** is axially slidably sleeved on the lower support rod **101** and above the positioning base **21**; and [0039] a stool assembly **3** with an upper end pivoted to the sliding base **22** and a lower end pivoted to the positioning base **21** via the collapsible rods **23**, wherein a bottom surface of a lower end of the stool assembly **3** is provided with first snap-fit portions **31**, the positioning base **21** is provided with a connector **4** which is provided with second snap-fit portions **41**, and when the stool assembly **3** is folded, the first snap-fit portions **31** are snap-fitted with the second snap-fit portions **41**.

[0040] According to the foldable crutch stool provided by the present invention, the stool assembly **3** in a folded state is reliably connected to the positioning base, such that the stool assembly **3** in the folded state is unlikely to be unfolded accidentally, which can improve the users' use experience, avoid accidents caused by accidentally unfolding the stool assembly **3**, and facilitate improvement of safety. Meanwhile, the foldable crutch stool is adjustable in height, which can thus meet the use requirements of more different users to some extent, presenting a wider range of application.

[0041] In this embodiment, the stool assembly **3** includes a middle seat plate **32** and side seat plates **33** disposed on both sides of the middle seat plate **32**, the side seat plates **33** are connected to the middle seat plate **32** in a foldable way, and the bottom surfaces on lower ends of the middle seat plate **32** and the side seat plates **33** are provided with the first snap-fit portions **31**. In this way, with the middle seat plate **32** and the side seat plates **33** on both sides, the folded stool assembly **3** can achieve an extremely small footprint, and can be substantially packaged outside the support rod **11**, which facilitates storage and carrying by a user. Furthermore, the bottom surfaces on the lower ends of the middle seat plate **32** and the side seat plates **33** are provided with the first snap-fit portions **31**. That is, each seat plate can be connected to the connector **4**, which further improves the reliability of connection between the folded stool assembly **3** and the positioning base **21**.

[0042] Meanwhile, in this embodiment, the support assembly **1** is adjustable in structural height and relatively convenient to operate and use. Specifically, when the lower support rod **101** and the upper support rod **102** need to be connected together, the connecting member **103** is sleeved outside the lower support rod **101**, and the first regulating hole **111** and the second regulating hole **121** corresponding to the first regulating hole in position are connected together via the connecting member **103**, thereby achieving connection between the lower support rod **101** and the upper support rod **102**. When the height of the support assembly **1** needs to be regulated again, the connecting member **103** is removed from the lower support rod **101**, the first regulating hole **111** and the second regulating hole **121** corresponding to the first regulating hole in position are disconnected by means of the connecting member **103**, in which case the upper support rod **102** can move vertically inside the lower support rod **101**, and when the upper support rod **102** moves to an appropriate position, the operation of connecting the lower support rod **101** and the upper support rod **102** is repeated.

[0043] In this embodiment, the connecting member **103** includes an elastic snap ring **131** and a connecting pin **132** disposed on an inner side of the elastic snap ring **131**, the elastic snap ring **131** is sleeveable outside the lower support rod **101**, and the connecting pin **132** is inserted into the first regulating hole **111** and the corresponding second regulating hole **121**. In this way, the connecting member **103** is simple in structure and low in cost, which facilitates batch production and market popularization. Meanwhile, the connecting member **103** is reasonable and ingenious in structure, by which the connection between the lower support rod **101** and the upper support rod **102** can be achieved, thereby achieving better convenience in operation and use.

[0044] In this embodiment, the elastic snap ring **131** is provided with a press portion **133**. In this way, the convenience in installation and use of the connecting member **103** can be improved.

[0045] Further, the connector **4** is provided with three second snap-fit portions **41** to correspondingly mate three first snap-fit portions **31** on the stool assembly **3**, exhibiting a simple structure and a skillful design. That is, the connector **4** can be connected to each seat plate, such that the reliability in connection of the stool assembly **3** in the folded state can be improved.

[0046] More further, the connector **4** is axially slidably sleeved on the support rod **11** and between the positioning base **21** and the sliding base **22**, the connector **4** is provided with connecting rods **42** connected to the positioning base **21**, and springs **5** are sleeved outside the connecting rods **42** and sandwiched between the connector **4** and the positioning base **21**. In this way, the structure is simple, and the design is reasonable and skillful, which facilitate the implementation of folding and unfolding of the crutch stool. Of course, a switching mechanism can be provided to implement the folding and unfolding of the crutch stool, which, however, pertains to an existing design and will not be described in detail.

[0047] In this embodiment, two connecting rods **42** are provided and located on either side of the connector **4** respectively, and the lower ends of the connecting rods **42** are provided with hooks **421** for connection with the positioning base **21**. In this way, the structure is simple, and the design is reasonable, which can improve the stability and reliability of connection between the connector **4** and the positioning base **21** to avoid looseness therebetween.

[0048] Further, the positioning base **21** is provided with a connecting base **6** at bottom, and the lower end of the stool assembly **3** is pivoted to the connecting base **6** via the collapsible rods **23**. In this way, the structure is simple, and the design is reasonable, which facilitate the improvement of stability and reliability of the crutch stool in use.

[0049] Further, the lower support rod **101** and the upper support rod **102** are both polygonal in cross section. The polygonal design improves the overall structural strength of the lower support rod **101** and the upper support rod **102**, such that the purpose of improving the stability and reliability of the crutch stool is achieved.

[0050] More further, the handle assembly **104** is 7-shaped and includes a transversely disposed hand gripping portion **141** and a connecting portion **142** extending from the bottom of the hand gripping portion **141**, and the connecting portion **142** is inserted into the upper support rod **102** to achieve their connection. In this way, the handle assembly **104** is simple in structure and reasonable and skillful in design, which facilitates the connection between the handle assembly **104** and the upper support rod **102** as well as the gripping by a user, thereby improving the user's use experience.

[0051] In this embodiment, the peripheral surface of the connecting portion **142** is provided with reinforcing ribs **1421** that are circumferentially distributed at interval, and the reinforcing ribs **1421** are axially arranged. In this way, the structure is simple, the design is reasonable, and the reinforcing ribs **1421** enhance the connection between the connecting portion **142** and the upper support rod **102**, which improves the stability and reliability in connection and avoids looseness between the two.

[0052] Further, a decorative ring **105** is sleeved on the top of the lower support rod **101**, and the upper support rod **102** is inserted into the decorative ring **105**. With the decorative ring **105** provided, the gap between the lower support rod **101** and the upper support rod **102** can be covered, which endows the crutch stool with an aesthetic appearance to improve the appeal to consumers and market competitiveness for the crutch stool.

[0053] More further, the lower support rod **101** is provided with at least two foldable support legs **106**, which are provided with anti-skid pads **107** at bottom. In this way, the structure is simple, and the foldable crutch stool in an unfolded state can be better supported on the ground by means of the support legs **106**, such that the sitting stability is improved for a user. Moreover, the support legs **106** have a foldable design, by which the folded crutch stool is smear in footprint to facilitate storage and carrying by the user after folding. Moreover, with the design of the anti-skid pads **107**, the anti-skid property of the crutch stool can be improved to prevent a user from slippage and

falling when he/she is sitting on the crutch stool.

[0054] Described above are only some embodiments of the present invention. For those of ordinary skill in the art, several variations and improvements can also be made without departing from the inventive concept of the present invention, and these variations and improvements shall be construed as falling within the protection scope of the present invention.

## Claims

1. A foldable crutch stool, comprising: a support assembly (1), comprising an axially disposed support rod (11), wherein the support rod (11) comprises an upper support rod (101) and a lower support rod (102), the lower support rod (101) is provided with a first regulating hole (111) close to an upper end of the lower support rod, and the upper support rod (102) is vertically movably disposed in the lower support rod (101), and is provided with several second regulating holes (121) mated with the first regulating hole; a connecting member (103) sleeved outside the lower support rod (101) for connecting the first regulating hole (111) and the second regulating hole (121) corresponding thereto in position, together; a handle assembly (104) disposed on a top of the upper support rod (102); a collapsing assembly (2), comprising a positioning base (21), a sliding base (22) and collapsible rods (23), wherein the positioning base (21) is sleeved on the lower support rod (101), and the sliding base (22) is axially slidably sleeved on the lower support rod (101) and above the positioning base (21); and a stool assembly (3) with an upper end pivoted to the sliding base (22) and a lower end pivoted to the positioning base (21) via the collapsible rods (23), wherein a bottom surface of a lower end of the stool assembly (3) is provided with first snap-fit portions (31), the positioning base (21) is provided with a connector (4) which is provided with second snap-fit portions (41), and when the stool assembly (3) is folded, the first snap-fit portions (31) are snap-fitted with the second snap-fit portions (41).
2. The foldable crutch stool according to claim 1, wherein the stool assembly (3) comprises a middle seat plate (32) and side seat plates (33) disposed on both sides of the middle seat plate (32), the side seat plates (33) are connected to the middle seat plate (32) in a foldable way, and bottom surfaces on lower ends of the middle seat plate (32) and the side seat plates (33) are provided with the first snap-fit portions (31).
3. The foldable crutch stool according to claim 2, wherein the connector (4) is provided with three second snap-fit portions (41) to correspondingly mate three first snap-fit portions (31) on the stool assembly (3).
4. The foldable crutch stool according to claim 1, wherein the connector (4) is axially slidably sleeved on the lower support rod (101) and between the positioning base (21) and the sliding base (22), the connector (4) is provided with connecting rods (42) connected to the positioning base (21), and springs (5) are sleeved outside the connecting rods (42) and sandwiched between the connector (4) and the positioning base (21).
5. The foldable crutch stool according to claim 4, wherein two connecting rods (42) are provided and located on either side of the connector (4) respectively, and lower ends of the connecting rods (42) are provided with hooks (421) for connection with the positioning base (21).
6. The foldable crutch stool according to claim 1, wherein the positioning base (21) is provided with a connecting base (6) at bottom, a lower end of the stool assembly (3) is pivoted to the connecting base (6) via the collapsible rods (23).
7. The foldable crutch stool according to claim 1, wherein the connecting member (103) comprises an elastic snap ring (131) and a connecting pin (132) disposed on an inner side of the elastic snap ring (131), the elastic snap ring (131) is sleeveable outside the lower support rod (101), the connecting pin (132) is inserted into the first regulating hole (101) and the corresponding second regulating hole (121), and the elastic snap ring (131) is provided with a press portion (133).
8. The foldable crutch stool according to claim 1, wherein the lower support rod (101) and the

upper support rod (102) are both polygonal in cross section, the handle assembly (104) is 7-shaped and comprises a transversely disposed hand gripping portion (141) and a connecting portion (142) extending from a bottom of the hand gripping portion (141), the connecting portion (142) is inserted into the upper support rod (102) to achieve connection, a peripheral surface of the connecting portion (142) is provided with reinforcing ribs (1421) that are circumferentially distributed at interval, and the reinforcing ribs (1421) are axially arranged.

**9.** The foldable crutch stool according to claim 1, wherein a decorative ring (105) is sleeved on a top of the lower support rod (101), and the upper support rod (102) is inserted into the decorative ring (105).

**10.** The foldable crutch stool according to claim 1, wherein the lower support rod (101) is provided with at least two foldable support legs (106), which are provided with anti-skid pads (107) at bottom.

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